Zihao He

Email   Homepage   Google Scholar

RESEARCH INTERESTS

RLHF, Reward Modeling, LLM Alignment & Safety, LLM Evaluation, Vision Language Models

EXPERIENCE

Meta — **Menlo Park, CA**
Research Scientist, Monetization GenAI — *02/2025 – Present*

- Train reward models on large-scale pairwise human preference data for image and video generation, including tie-aware Bradley-Terry modeling, calibration analysis, and model–human agreement optimization.
- Design and run RLHF pipelines that use trained reward models to steer generative model behavior via policy gradient methods, improving alignment with human quality judgments across fine-grained visual dimensions.
- Build LLM-as-a-Judge automated evaluation systems for multimodal generation, defining rubrics that decompose quality into interpretable sub-dimensions and correlate with human ratings.
- Lead multimodal data curation pipelines — prompt rewriting, synthetic augmentation, bias detection, debiasing — to prevent systematic artifacts in reward model training data.
- Develop scalable distributed training and inference workflows (multi-node, multi-GPU) for large vision-language reward and policy models.
- Ship production reward and evaluation tools that directly inform model iteration and release gating, resulting in measurable improvements in generative quality.

TikTok Inc. — **San Jose, CA**
Research Scientist Intern — *05/2023 – 08/2023*

- Developed alignment-based music captioning via contrastive audio-text pretraining (published at EMNLP 2023).

Amazon.com Services, Inc. — **Seattle, WA (Remote)**
Applied Scientist Intern — *05/2021 – 08/2021*

- Built text-embedding-based book personalization for Amazon Kids.

DiDi Global — **Beijing, China**
Algorithm Development Intern — *04/2019 – 07/2019*

- Multi-task learning for multi-label driver misbehavior classification.

EDUCATION

University of Southern California — **Los Angeles, CA**
Ph.D. in Computer Science — *08/2019 – 02/2025*

- Advisor: Prof. Kristina Lerman   Thesis: *Aligning Large Language Models with Human Perspectives*

Tsinghua University — **Beijing, China**
M.Eng. in Computer Technologies (Transferred) — *09/2018 – 06/2019*

Beijing University of Posts and Telecommunications — **Beijing, China**
B.Eng. in Communications Engineering — *09/2014 – 06/2018*

SELECTED PUBLICATIONS

- J. Wu, X. Kan, **Z. He**, S. Tan, B. Pan, K. Zhang. "Multi-Task Reinforcement Learning for Enhanced Multimodal LLM-as-a-Judge." *ACL Industry Track*, 2026. [Link]
- **Z. He**, S. Guo, A. Rao, K. Lerman. "Whose Emotions and Moral Sentiments Do Language Models Reflect?" *Findings of ACL*, 2024. [Link]
- **Z. He**, M.D. Chu, R. Dorn, S. Guo, K. Lerman. "Community-Cross-Instruct: Unsupervised Instruction Generation for Aligning Large Language Models to Online Communities." *EMNLP*, 2024. [Link]
- K. Chen, **Z. He**, J. Yan, T. Shi, K. Lerman. "How Susceptible are Large Language Models to Ideological Manipulation?" *EMNLP*, 2024. [Link] Best Paper Runner-up, SeT LLM @ ICLR 2024
- M.D. Chu, **Z. He**, R. Dorn, K. Lerman. "Improving and Assessing the Fidelity of LLM Alignment to Online Communities." *NAACL*, 2025. [Link]
- K. Chen, T. Shi, **Z. He**, K. Lerman. "Steer-Bench: A Benchmark for Evaluating the Steerability of Large Language Models." *EMNLP*, 2025. [Link]
- **Z. He**, N. Mokhberian, K. Lerman. "Infusing Knowledge from Wikipedia to Enhance Stance Detection." *WASSA @ EMNLP*, 2022. [Link]
- **Z. He**, L. Tavabi, K. Lerman, M. Soleymani. "Speaker Turn Modeling for Dialogue Act Classification." *Findings of EMNLP*, 2021. [Link]
- **Z. He**, W. Hao, W.T. Lu, C. Chen, K. Lerman, X. Song. "ALCAP: Alignment-Augmented Music Captioner." *EMNLP*, 2023. [Link]
- H. Nguyen, **Z. He**, et al. "Smoothing Out Hallucinations: Mitigating LLM Hallucination with Smoothed Knowledge Distillation." *arXiv*, 2025. [Link]
- **Z. He**, J. May, K. Lerman. "CPL-NoViD: Context-aware Prompt-based Learning for Norm Violation Detection in Online Communities." *ICWSM*, 2024. [Link]

HONORS

- Best Paper Runner-up, SeT LLM @ ICLR 2024
- Annenberg Fellowship, University of Southern California, 2019
- National Scholarship (top 1.2%), Beijing University of Posts and Telecommunications, 2018

PROFESSIONAL SERVICE

Reviewer: ACL Rolling Review (11 cycles, 2023–2026), COLING 2024/2025, ICWSM 2024, TrustNLP @ NAACL/ACL 2024–2026, R2HCAI @ AAAI 2023. Program Committee: CCS 2021.

SKILLS

Languages	Python, C++, CUDA
ML / RL	PyTorch, Transformers, DeepSpeed, FSDP, vLLM, TRL
Data	NumPy, Pandas, Scikit-Learn, Weights & Biases